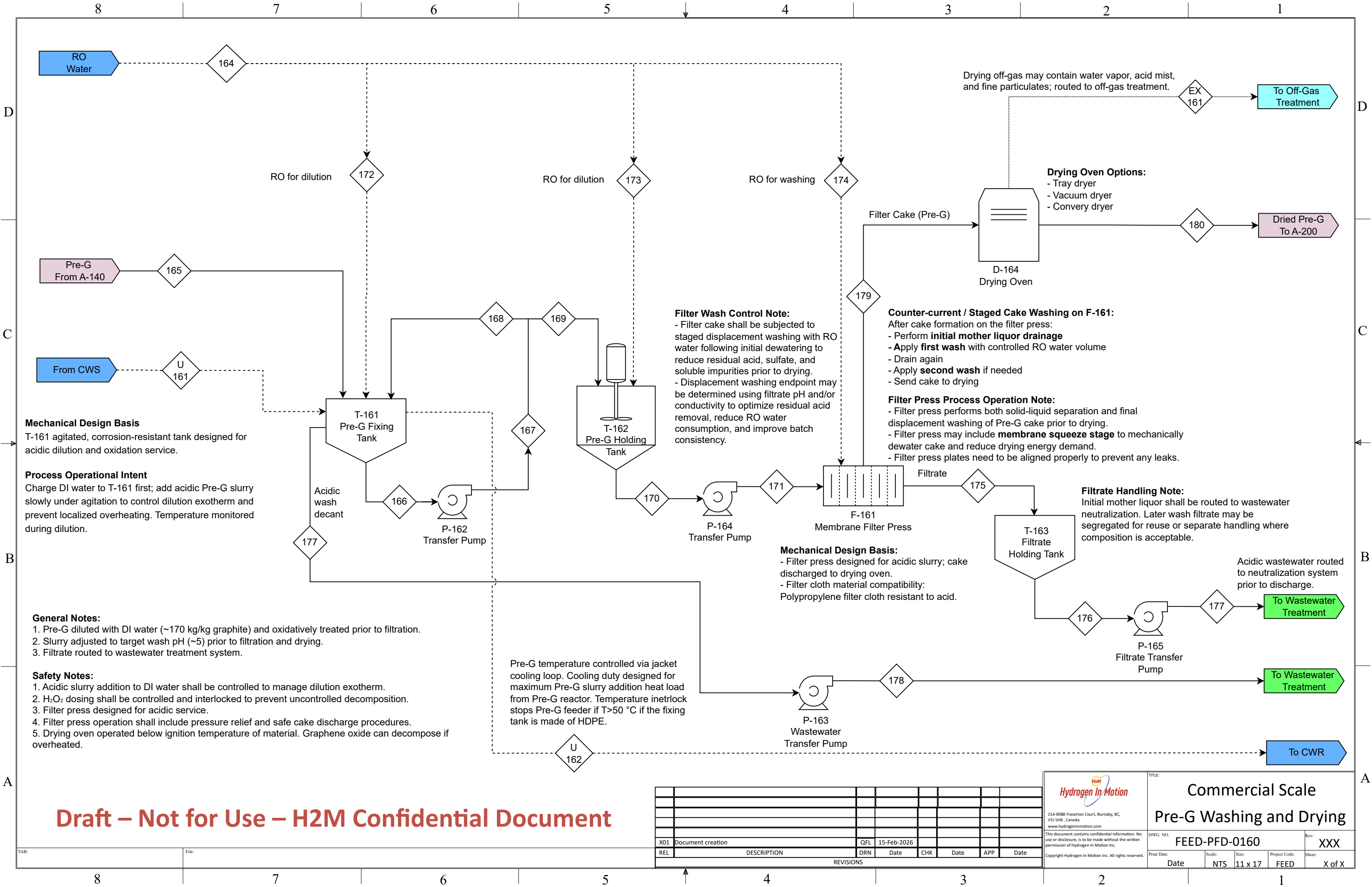
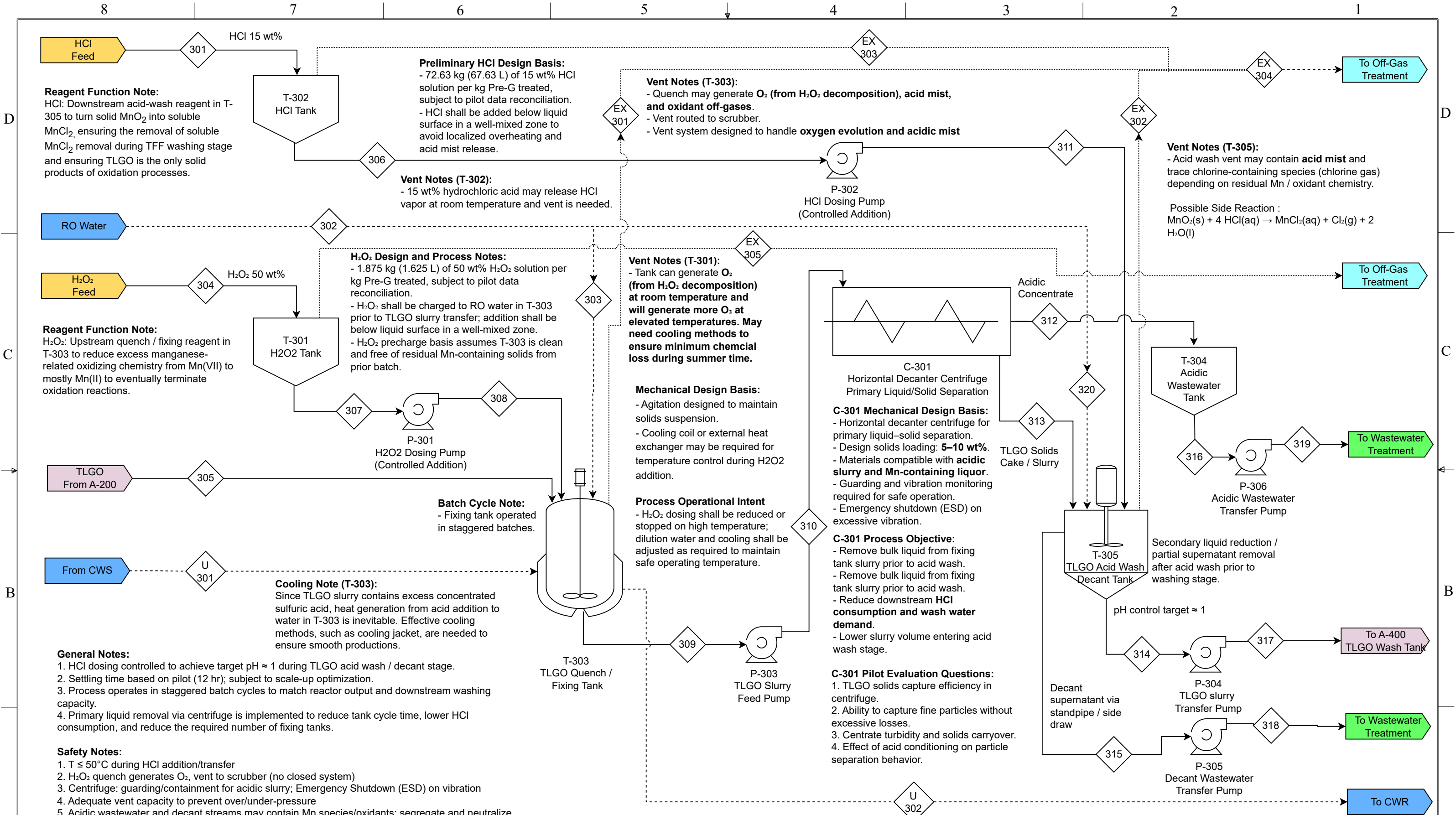










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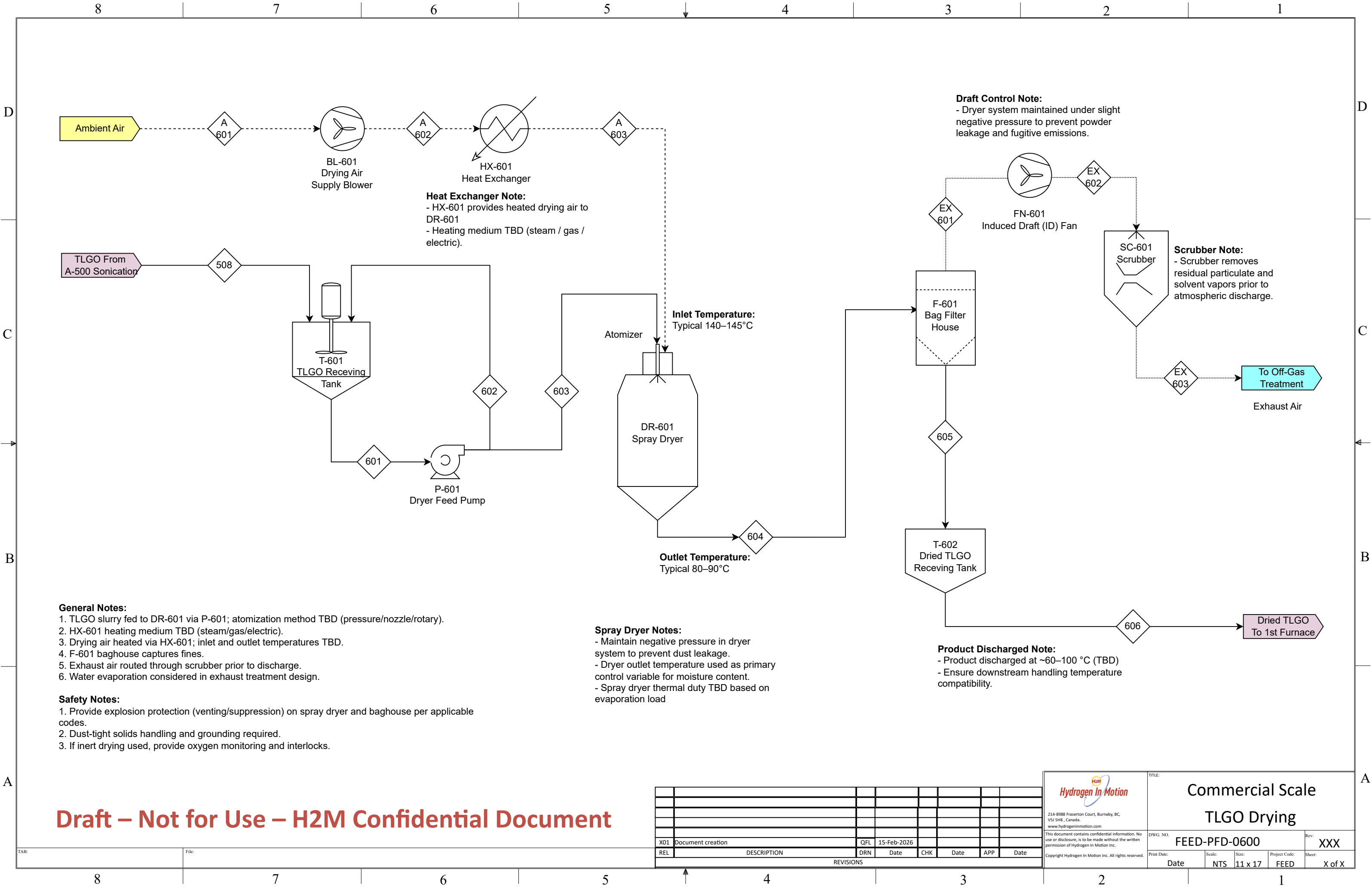
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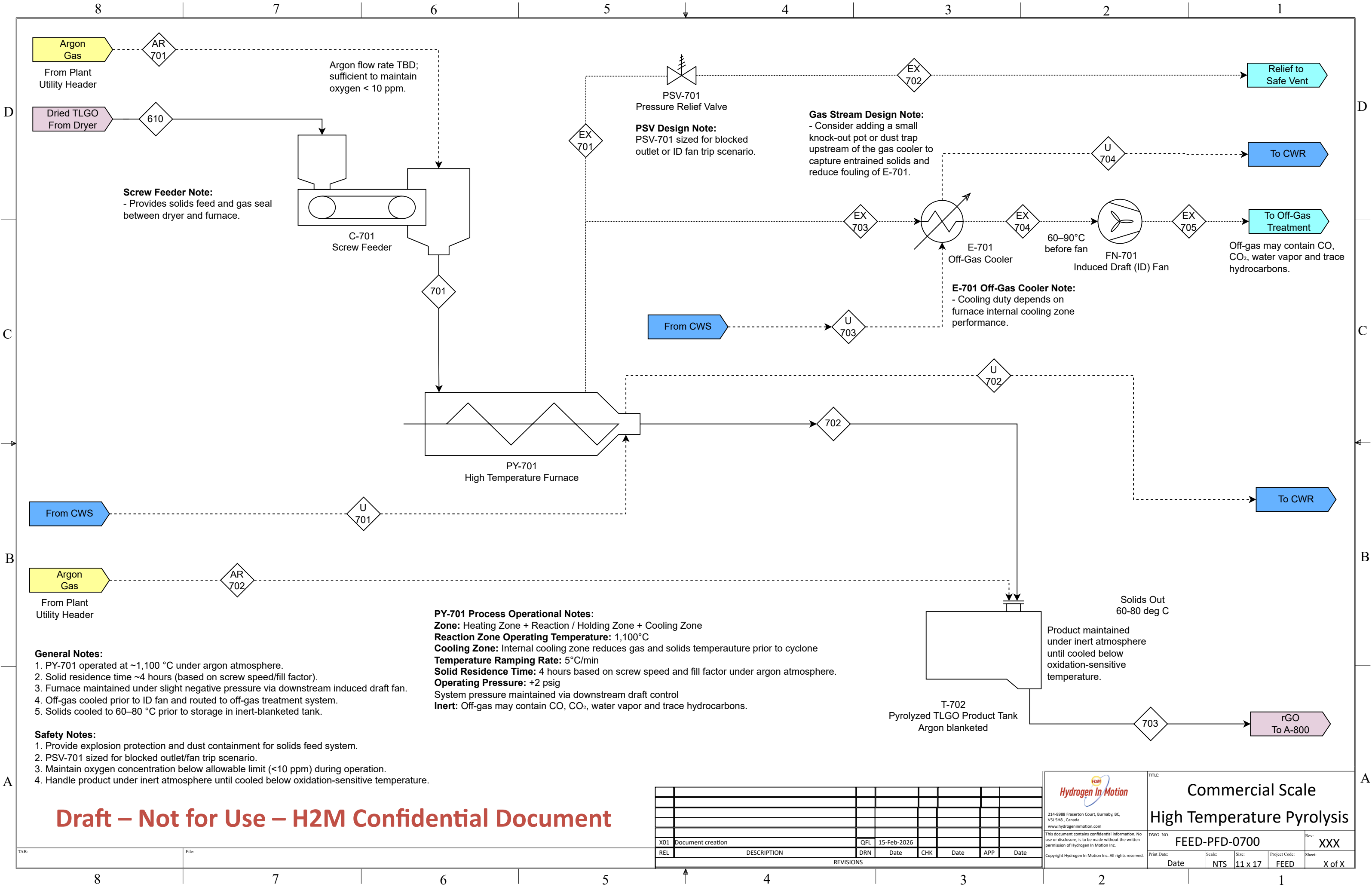
DWG. NO. **FEED-PFD-0300**

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TITLE:

Commercial Scale High Temperature Pyrolysis

DWG. NO.

FEED-PFD-0700

Rev:

XXX

Print Date:

Date

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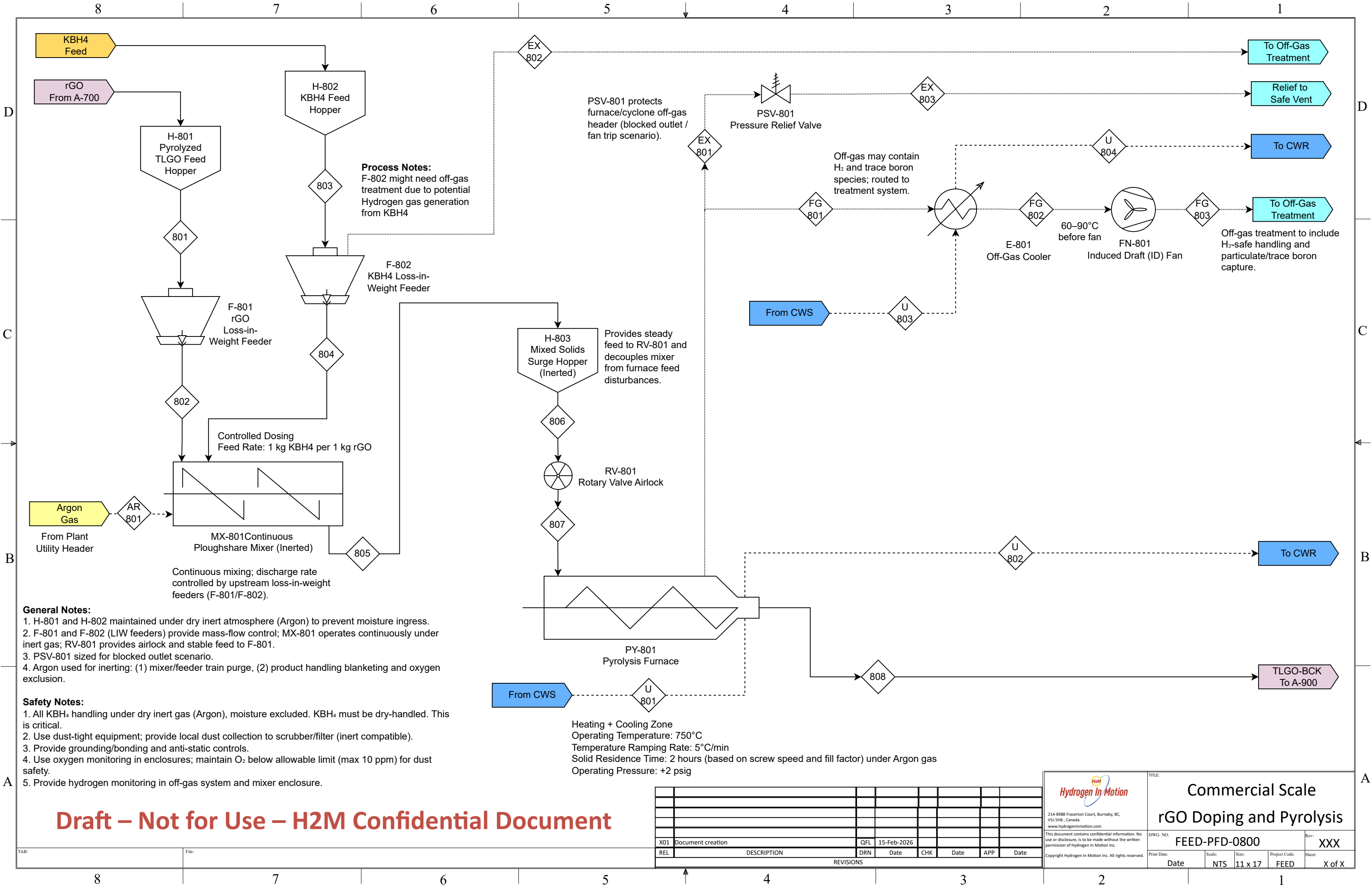
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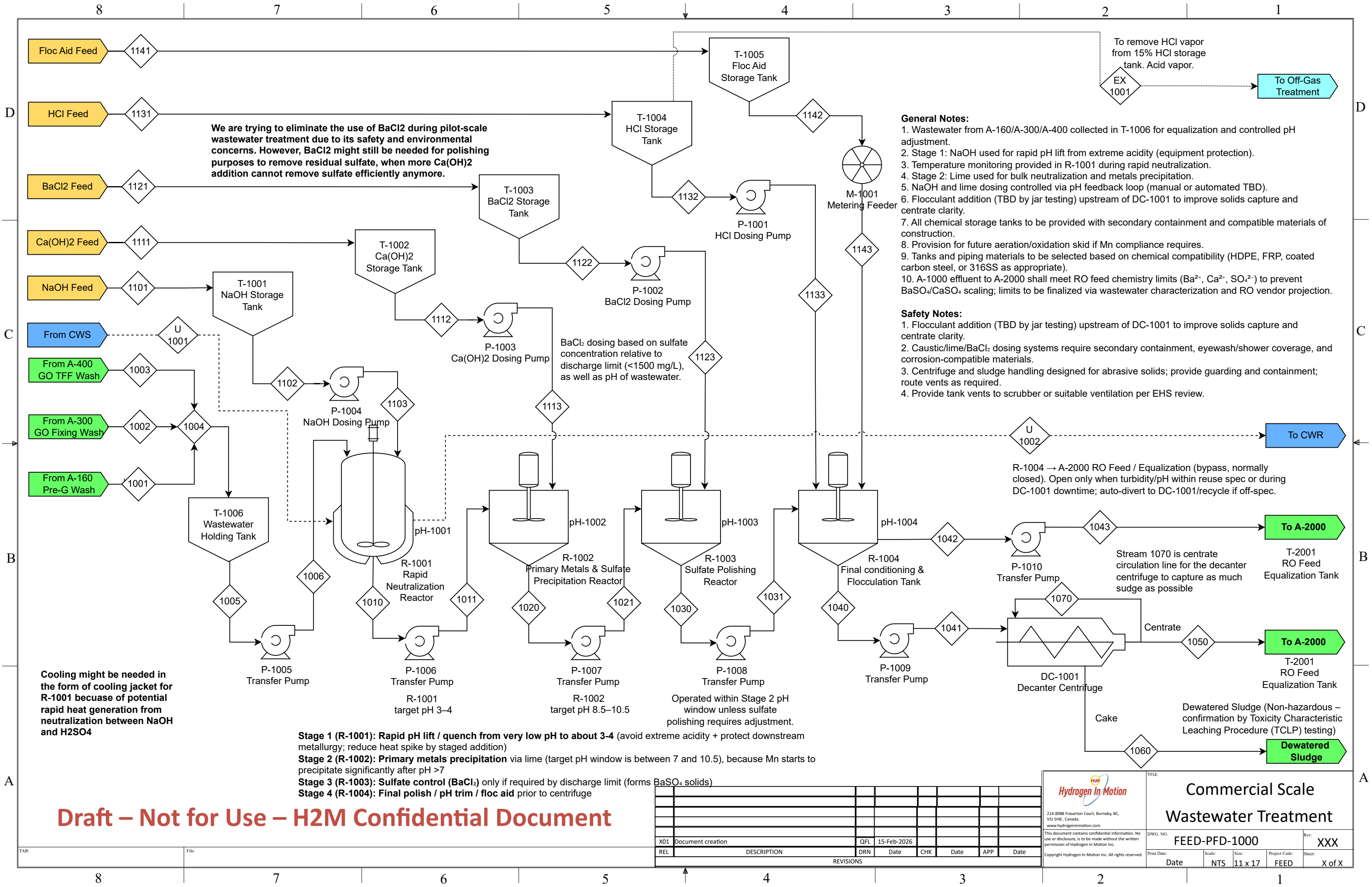
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TITLE: **Commercial Scale
rGO Doping and Pyrolysis**

DWG. NO. **FEED-PFD-0800**

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We are trying to eliminate the use of BaCl₂ during pilot-scale wastewater treatment due to its safety and environmental concerns. However, BaCl₂ might still be needed for polishing purposes to remove residual sulfate, when more Ca(OH)₂ addition cannot remove sulfate efficiently anymore.

- General Notes:**
- 1. Wastewater from A-160/A-300/A-400 collected in T-1006 for equalization and controlled pH adjustment.
 - 2. Stage 1: NaOH used for rapid pH lift from extreme acidity (equipment protection).
 - 3. Temperature monitoring provided in R-1001 during rapid neutralization.
 - 4. Stage 2: Lime used for bulk neutralization and metals precipitation.
 - 5. NaOH and lime dosing controlled via pH feedback loop (manual or automated TBD).
 - 6. Flocculant addition (TBD by jar testing) upstream of DC-1001 to improve solids capture and centrate clarity.
 - 7. All chemical storage tanks to be provided with secondary containment and compatible materials of construction.
 - 8. Provision for future aeration/oxidation skid if Mn compliance requires.
 - 9. Tanks and piping materials to be selected based on chemical compatibility (HDPE, FRP, coated carbon steel, or 316SS as appropriate).
 - 10. A-1000 effluent to A-2000 shall meet RO feed chemistry limits (Ba²⁺, Ca²⁺, SO₄²⁻) to prevent BaSO₄/CaSO₄ scaling; limits to be finalized via wastewater characterization and RO vendor projection.
- Safety Notes:**
- 1. Flocculant addition (TBD by jar testing) upstream of DC-1001 to improve solids capture and centrate clarity.
 - 2. Caustic/lime/BaCl₂ dosing systems require secondary containment, eyewash/shower coverage, and corrosion-compatible materials.
 - 3. Centrifuge and sludge handling designed for abrasive solids; provide guarding and containment; route vents as required.
 - 4. Provide tank vents to scrubber or suitable ventilation per EHS review.

- Stage 1 (R-1001):** Rapid pH lift / quench from very low pH to about 3-4 (avoid extreme acidity + protect downstream metallurgy; reduce heat spike by staged addition)
- Stage 2 (R-1002):** Primary metals precipitation via lime (target pH window is between 7 and 10.5), because Mn starts to precipitate significantly after pH >7
- Stage 3 (R-1003):** Sulfate control (BaCl₂) only if required by discharge limit (forms BaSO₄ solids)
- Stage 4 (R-1004):** Final polish / pH trim / floc aid prior to centrifuge

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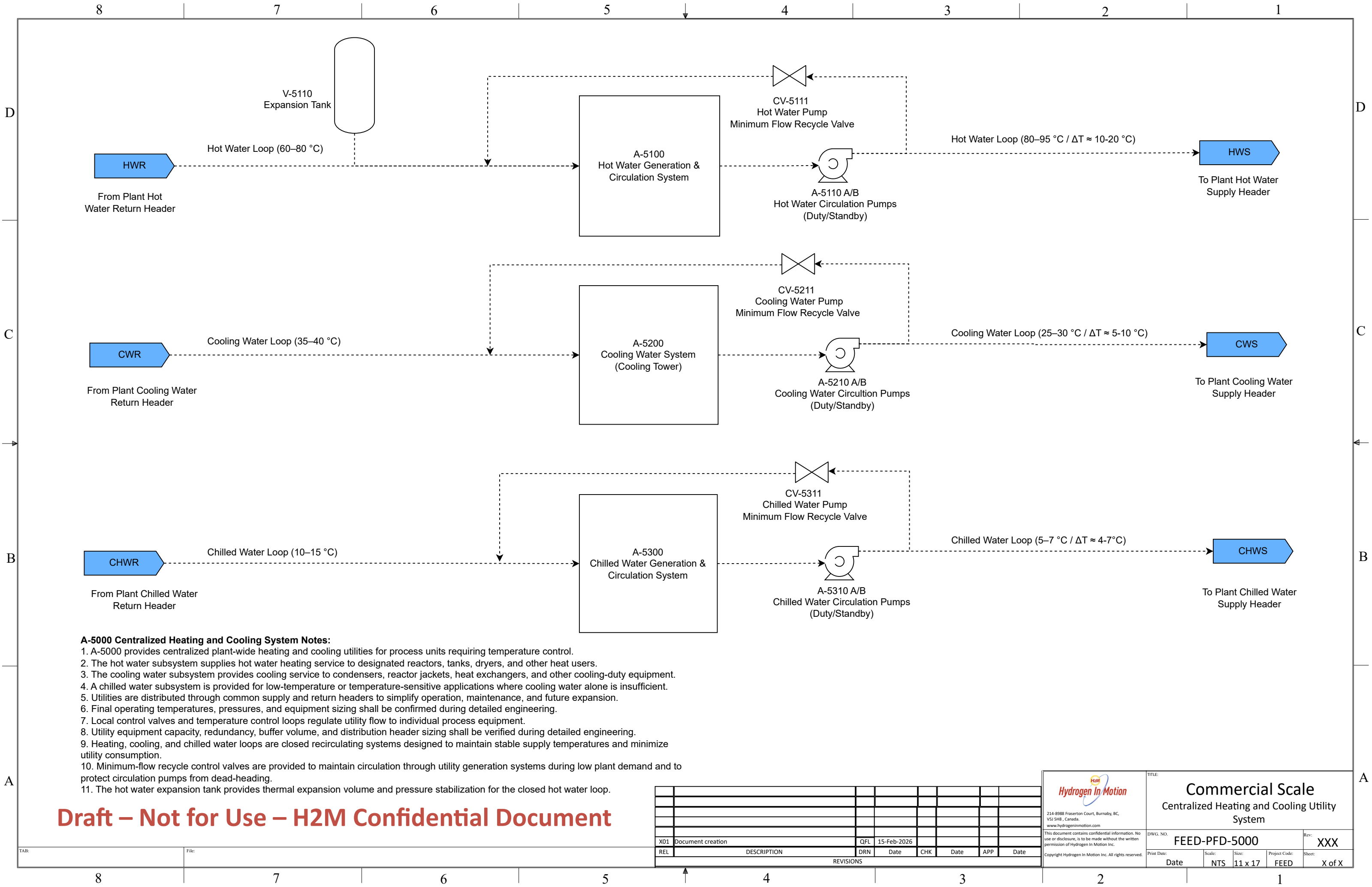
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TITLE: **Commercial Scale Wastewater Treatment**

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A-5000 Centralized Heating and Cooling System Notes:

- 1. A-5000 provides centralized plant-wide heating and cooling utilities for process units requiring temperature control.
- 2. The hot water subsystem supplies hot water heating service to designated reactors, tanks, dryers, and other heat users.
- 3. The cooling water subsystem provides cooling service to condensers, reactor jackets, heat exchangers, and other cooling-duty equipment.
- 4. A chilled water subsystem is provided for low-temperature or temperature-sensitive applications where cooling water alone is insufficient.
- 5. Utilities are distributed through common supply and return headers to simplify operation, maintenance, and future expansion.
- 6. Final operating temperatures, pressures, and equipment sizing shall be confirmed during detailed engineering.
- 7. Local control valves and temperature control loops regulate utility flow to individual process equipment.
- 8. Utility equipment capacity, redundancy, buffer volume, and distribution header sizing shall be verified during detailed engineering.
- 9. Heating, cooling, and chilled water loops are closed recirculating systems designed to maintain stable supply temperatures and minimize utility consumption.
- 10. Minimum-flow recycle control valves are provided to maintain circulation through utility generation systems during low plant demand and to protect circulation pumps from dead-heading.
- 11. The hot water expansion tank provides thermal expansion volume and pressure stabilization for the closed hot water loop.

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